

Household Bills — Android App

A private utility-bills companion that helps you understand spending and spot deviations.

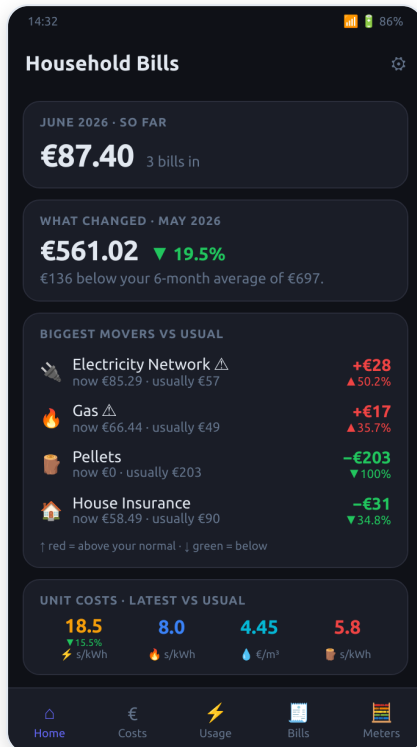
Version 1.0 · June 2026 · Single device (Galaxy S25) · Built & signed on a home NAS

What it is

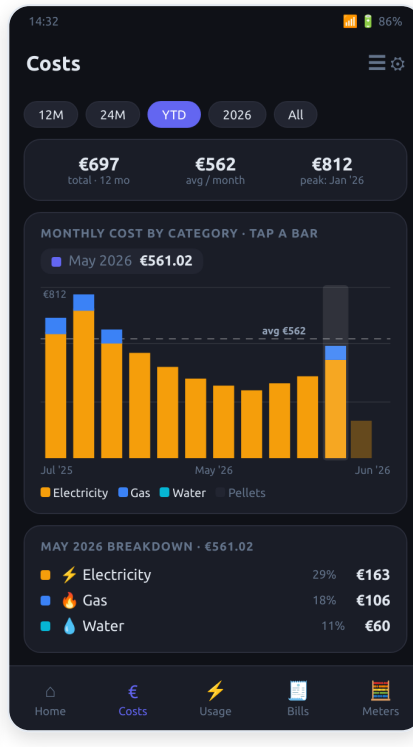
A native Android app that puts the home utility-bills dashboard on your phone — and goes beyond a read-only view to **help you make decisions**: it surfaces what changed, which categories moved away from their normal, and whether your unit rates are drifting. It is installed on one phone only, signs in with your Google account, and shows exactly the same numbers as the web dashboard because both read through one shared calculation engine.

[Kotlin · Jetpack Compose](#)[Material 3 \(dark\)](#)[Google Sign-In](#)[HTTPS only](#)[Offline cache](#)[Sideloaded APK](#)

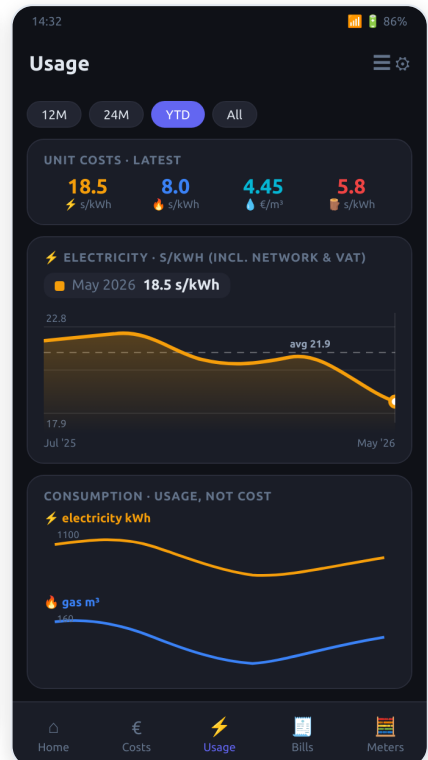
The screens



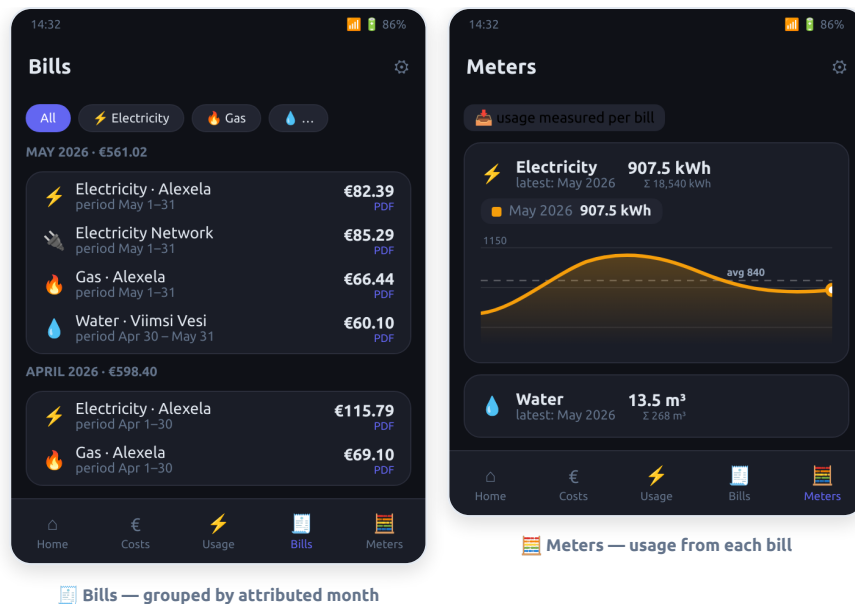
🏠 Home — what changed & biggest movers



€ Costs — bars, drill-down, range presets



⚡ Usage — unit costs with scale & average



The five tabs

TAB	WHAT IT SHOWS
Home	Decisions first. "This month so far"; <i>What changed</i> — the last complete month against its 6-month average with a plain-language verdict; <i>Biggest movers</i> — each category vs its own normal (red = above, green = below, Δ on genuine spikes); unit costs with a trend arrow; spending-by-category breakdown; and a 12-month trend line.
€ Costs	Monthly cost by category as stacked bars — tap any bar to drill into that month's full breakdown. Category breakdown (avg/month), year-over-year comparison, and a monthly-total line with a 3-month forecast (trailing 6-month average). Tap the legend to add or remove categories.
Usage	Unit costs front and centre: latest values plus a full history line for each of the five series — electricity s/kWh, gas s/kWh and s/m ³ , water €/m ³ , pellets s/kWh — each with its average reference line. Below: consumption trends (kWh / m ³), usage not cost.
Bills	Every bill grouped by month with subtotals, filterable by category. Bills are grouped by their attributed month (a transport bill issued in June for May service sits under May, matching the dashboard). Tap a row to open its original PDF.
Meters	Per-utility usage measured on each bill (kWh / m ³) — latest reading, cumulative total, and a history line. Reads the real per-period values from the bills.

Settings (account, server, sign out) sits behind the gear icon in the top bar.

Decision support — understanding deviations

The app's distinguishing feature. For the latest complete month it compares every category against its own **trailing six-month baseline** and tells you the story, not just the number:

- **What changed** — e.g. "May €561 — €136 below your 6-month average of €697 (▼19.5%)."
- **Biggest movers** — categories sorted by how far they moved from normal, with the euro delta and percentage, colour-coded by direction.
- **Anomaly flags (Δ)** — only genuine spikes are flagged (a category climbing well above its norm); a category dropping to zero out of season, like pellets in summer, is not treated as an alert.
- **Unit-cost trends** — each rate shown against its 6-month average, so a tariff drift is obvious at a glance.

Charts & interaction

Readable by design

Interactive

- Y-axis value labels (you can read the magnitude, not just the shape)
- A dashed *average* reference line on every chart, so deviations are visible against "normal"
- Smooth animated lines with gradient fills; rounded, animated bars
- Tap any point or bar → a guide line and a value pill (month + exact value)
- Tap a Costs bar → that month's category breakdown opens below
- Gaps where data is missing — never a misleading drop to zero

Filtering & analysis

One-tap range presets are always visible on Costs and Usage — no menus to dig through:

12M

24M

YTD

2026

2025

2024

All

A range-summary card shows the total, average per month, and peak month for whatever you've selected. The chart legend is tappable to include or exclude categories — the whole view re-slices instantly, on-device.

Security & access

- **Google Sign-In, one account only.** The server verifies the Google identity and accepts exactly one allowlisted email — any other account is refused.
- **HTTPS only** to the household server; the session token is stored in the Android Keystore (encrypted), never in logs or backups.
- **Fingerprint-ready** and single-device by design — no Play Store, no other users.

How the data flows

Utility bill email → Gmail scraper (daily 09:00) → parser → database

↓

bills_api —HTTPS :8443—> Android app (live on open / pull-to-refresh)

↑ shared calculation engine = identical numbers to the web dashboard

- **Freshness:** the app pulls live data every time you open it or pull to refresh; offline it shows the last sync with a clear banner.
- **New bills** are ingested once a day at 09:00; a Telegram summary is sent on each run, so you are notified when something new arrives.
- **Parity:** cost-month attribution and the accrual clip (e.g. annual insurance spread, pellets by heating season) live in one shared module used by both the app and the dashboard.

Under the hood

AREA	DETAIL
App	Kotlin, Jetpack Compose, Material 3 dark; minSdk 33 , arm64; package ee.household.bills
Networking	Retrofit + kotlinx-serialization over HTTPS; session JWT in EncryptedSharedPreferences
Server	/api/auth/google , /api/bills/summary , /api/bills/series , /api/bills , /api/meter-readings
Build	Signed release APK built in a throttled container on the NAS; delivered to Telegram; sideloaded
Charts	Hand-built on Compose Canvas: animated, gradient-filled, scale-labelled, interactive

In one line: open the app and it tells you where this month stands, what moved, and whether your rates are drifting — the same trustworthy numbers as the dashboard, in a form built for a phone.